

01. pom.xml - WebFlux 依赖配置

=== pom.xml Day4 新增依赖: spring-boot-starter-webflux ===

```
<!-- ===== -->
<!-- Day4 新增: WebFlux (仅使用 WebClient 调用 Python AI) -->
<!-- ===== -->
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-webflux</artifactId>
</dependency>
```

=== 验证: mvn compile ===

[INFO] Compiling 38 source files with javac [debug release 17] to target\classes

[INFO] BUILD SUCCESS

[INFO] Total time: 7.155 s

[OK] WebFlux 依赖添加成功, 编译通过

02. application.yml - 50MB 与 AI 配置

=== application.yml Day4 新增配置段 ===

```
spring:
  servlet:
    multipart:
      max-file-size: 50MB
      max-request-size: 50MB

  campus-qa:
    upload:
      dir: ${UPLOAD_DIR:storage/uploads}

  ai:
    base-url: ${AI_BASE_URL:http://localhost:8000}
    connect-timeout: 5
    response-timeout: 120
    callback-token: ${AI_CALLBACK_TOKEN:day4-internal-callback-token}
```

[OK] 50MB 上传限制已配置

[OK] AI 服务地址、超时、回调 Token 已配置

[OK] 所有敏感值使用环境变量占位

03. mvn clean compile - BUILD SUCCESS

=== mvn clean compile ===

```
[INFO] Scanning for projects...
[INFO] -----< com.group9:campus-qa >-----
[INFO] Building Campus QA Assistant 0.1.0
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ campus-qa ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ campus-qa ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 38 source files with javac [debug release 17] to target\classes
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 7.155 s
[INFO] Finished at: 2026-07-17T13:57:42+08:00
```

[OK] 38 个 Java 源文件全部编译成功

[OK] 新增 Day4 文件: KbDocument, KbDocumentMapper, DocumentQueryDTO,
DocumentUpdateDTO, DocumentVO, FileStorageUtil, AsyncConfig, WebClientConfig

04. ApplicationSmokeTest - 8/8 通过

```
=== mvn test -Dtest=ApplicationSmokeTest ===
```

```
[INFO] Running com.group9.campusqa.ApplicationSmokeTest
```

```

      _ _ _ _ _
     / _/   _/   _/   _/   _/
    ( )\  | ' _| ' _| ' _| ' _|
   V _|| _|| _|| _|| _|| _|| _||
   ' _|| _|| _|| _|| _|| _|| _||
   =====|_|=====|_|/_/_/_/_/
:: Spring Boot ::                (v3.2.7)
```

```
13:57:57.001 INFO --- AsyncConfig : 文档处理线程池已初始化: core=2, max=4, queue=50
```

```
13:57:57.327 INFO --- WebClientConfig : AI WebClient 已初始化: baseUrl=http://localhost:8000, connectTimeout=5s, responseTim
```

```
13:57:58.074 INFO --- ApplicationSmokeTest : Started ApplicationSmokeTest in 2.956 seconds
```

```
[INFO] Tests run: 8, Failures: 0, Errors: 0, Skipped: 0
```

```
[INFO] BUILD SUCCESS
```

```
[OK] ApplicationSmokeTest 全部 8 个测试通过
```

```
[OK] AsyncConfig 线程池初始化成功 (core=2, max=4, queue=50)
```

```
[OK] WebClientConfig AI WebClient 初始化成功 (baseUrl=http://localhost:8000)
```

```
[OK] Spring Boot 3.2.7 启动正常
```

05. kb_document 表结构设计

=== kb_document 表结构 (sql/day4_kb_document.sql) ===

```
CREATE TABLE IF NOT EXISTS kb_document (
  id          BIGINT          AUTO_INCREMENT PRIMARY KEY COMMENT '文档ID',
  title       VARCHAR(255)    NOT NULL          COMMENT '展示标题',
  original_name VARCHAR(255)  NOT NULL          COMMENT '上传原文件名',
  stored_name  VARCHAR(255)    NOT NULL          COMMENT 'UUID存储名',
  file_path    VARCHAR(500)    NOT NULL          COMMENT '文件绝对路径',
  file_type    VARCHAR(20)     NOT NULL          COMMENT '文件类型: pdf/doc/docx/txt',
  file_size    BIGINT          NOT NULL          DEFAULT 0 COMMENT '文件字节数',
  chunk_count  INT             NOT NULL          DEFAULT 0 COMMENT '切分块数',
  status       VARCHAR(30)     NOT NULL          DEFAULT 'PENDING' COMMENT '处理状态',
  process_stage VARCHAR(30)    NULL             DEFAULT 'UPLOADED' COMMENT '当前处理阶段',
  error_message VARCHAR(1000)  NULL             COMMENT '失败原因',
  create_user_id BIGINT        NOT NULL          COMMENT '上传管理员ID',
  create_time  DATETIME        NOT NULL          DEFAULT CURRENT_TIMESTAMP,
  update_time  DATETIME        NOT NULL          DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  deleted      TINYINT         NOT NULL          DEFAULT 0 COMMENT '逻辑删除',
  -- 索引
  INDEX idx_kb_document_status (status),
  INDEX idx_kb_document_file_type (file_type),
  INDEX idx_kb_document_create_user_id (create_user_id),
  INDEX idx_kb_document_create_time (create_time),
  INDEX idx_kb_document_deleted (deleted)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

[OK] 15 个字段与文档约定完全一致

[OK] 5 个性能索引

[OK] 状态约束: PENDING/PROCESSING/COMPLETED/FAILED

06. GitHub PR #11 页面信息

== GitHub Pull Request #11 ==

[LINK] <https://github.com/WangJing-hhu/campus-practice-group9/pull/11>

分支: feat/day4-backend-doc-data → dev/day4

标题: feat: Day4 知识库文档数据模型与基础配置 (王雨淇)

审查者: 杨熙杰

修改文件 (13 个):

新增:

- sql/day4_kb_document.sql
- entity/KbDocument.java
- mapper/KbDocumentMapper.java
- dto/DocumentQueryDTO.java
- dto/DocumentUpdateDTO.java
- vo/DocumentVO.java
- util/FileStorageUtil.java
- config/AsyncConfig.java
- config/WebClientConfig.java
- Day4/王雨淇_个人完成说明.md
- Day4/王雨淇_日报.md

修改:

- backend/pom.xml (+WebFlux)
- application.yml (+50MB/AI配置)

[OK] PR 已创建, 等待杨熙杰审查

07. kb_document 建表验证 (DESC + SHOW INDEX)

=== kb_document 建表验证 ===

执行命令:

```
mysql -u root -p < sql/day4_kb_document.sql
```

表结构 (DESC kb_document):

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
title	varchar(255)	NO		NULL	
original_name	varchar(255)	NO		NULL	
stored_name	varchar(255)	NO		NULL	
file_path	varchar(500)	NO		NULL	
file_type	varchar(20)	NO	MUL	NULL	
file_size	bigint	NO		0	
chunk_count	int	NO		0	
status	varchar(30)	NO	MUL	PENDING	
process_stage	varchar(30)	YES		UPLOADED	
error_message	varchar(1000)	YES		NULL	
create_user_id	bigint	NO	MUL	NULL	
create_time	datetime	NO	MUL	CURRENT_TIMESTAMP	
update_time	datetime	NO		CURRENT_TIMESTAMP	on update CURRENT_TIMESTAMP
deleted	tinyint	NO	MUL	0	

索引验证 (SHOW INDEX FROM kb_document):

```
PRIMARY KEY (id)
idx_kb_document_status (status)
idx_kb_document_file_type (file_type)
idx_kb_document_create_user_id (create_user_id)
idx_kb_document_create_time (create_time)
idx_kb_document_deleted (deleted)
```

[OK] 15 个字段全部正确

[OK] 5 个索引全部创建

[OK] utf8mb4_unicode_ci 字符集

[OK] InnoDB 引擎

08. Spring Boot 8081 启动验证

=== Spring Boot 8081 启动验证 (连接 MySQL 端口 3307) ===

```
mvn spring-boot:run (DB_PORT=3307 DB_USERNAME=root DB_PASSWORD=root123)
```

```
14:04:32.480 INFO CampusQaApplication : Starting CampusQaApplication using Java 17.0.19
```

```
14:04:33.467 INFO TomcatWebServer : Tomcat initialized with port 8081 (http)
```

14:04:33.949 INFO AsyncConfig : 文档处理线程池已初始化: core=2, max=4, queue=50

```
14:04:34.094 INFO WebClientConfig : AI WebClient 已初始化: baseUrl=http://localhost:8000,
connectTimeout=5s, responseTimeout=120s
```

```
14:04:34.341 INFO TomcatWebServer : Tomcat started on port 8081 (http)
```

```
14:04:34.347 INFO CampusQaApplication : Started CampusQaApplication in 2.194 seconds
```

说明: 接口正常响应, JWT 认证保护生效 [OK]

[OK] WebClientConfig AI 服务客户端初始化成功

[OK] /api/doc/** 接口受 JWT 保护

[OK] Day4 新增 Bean 全部正常加载